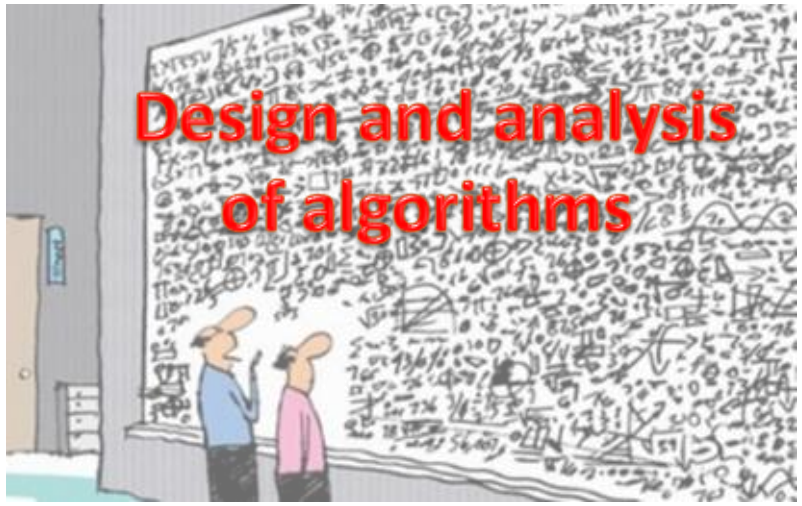


- **Perfect hashing**

The problem

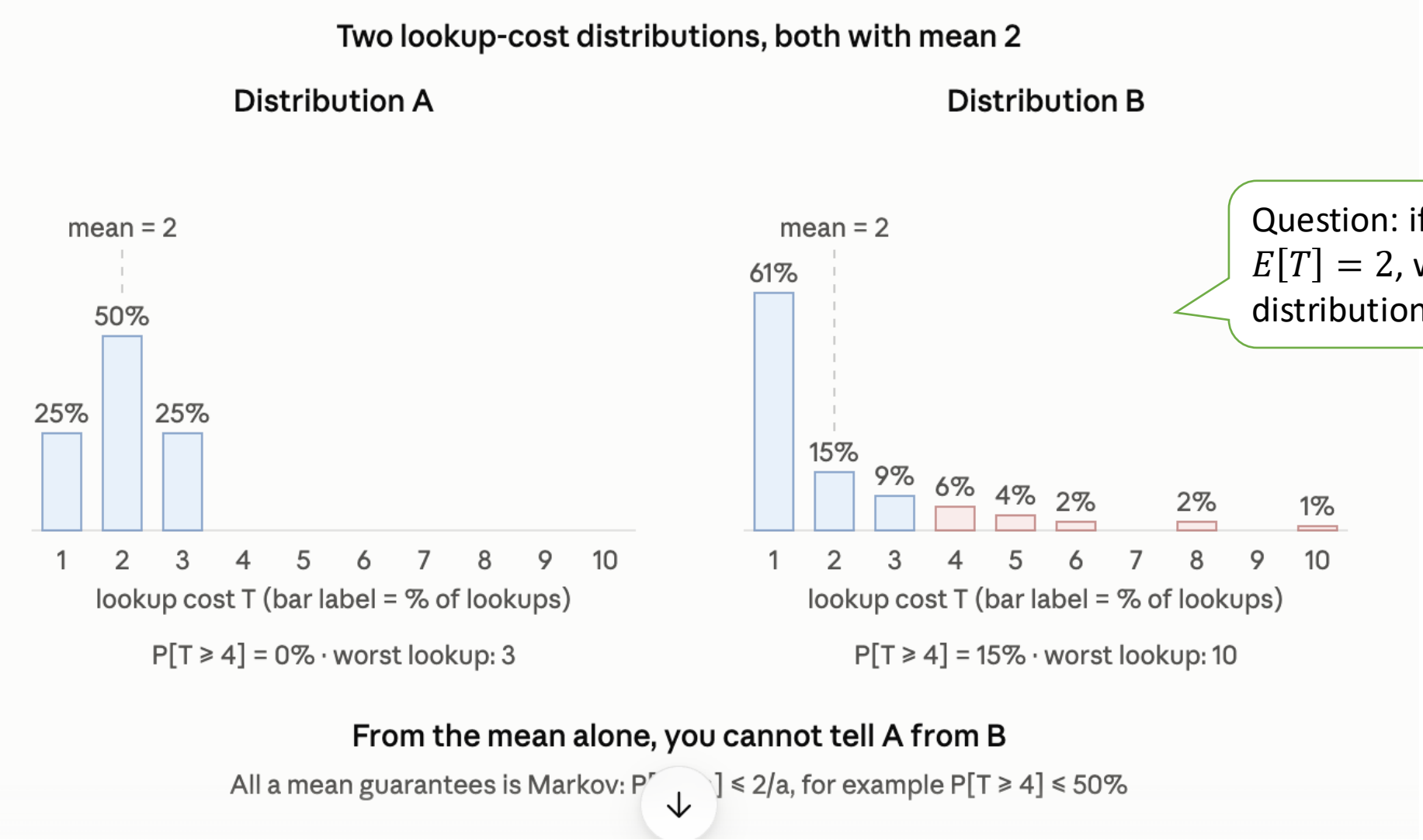
say using universal hash functions

- N keys, hash table size M.
- We know that using chaining, the **expected time** $E[T]$ for $\text{find}(x)$ is $O\left(\frac{N}{M}\right) = O(1)$ if space $M = \Theta(N)$.
- Can we make the **actual time** $\boldsymbol{T = O(1)}$ using similar $O(N)$ space?
- Not really! *if the set of keys is arbitrary*, it is like throwing balls in bins…
  - Nobody can beat laws of probability.
- Hence, we need a special case of the problem.
- Let's understand this more!

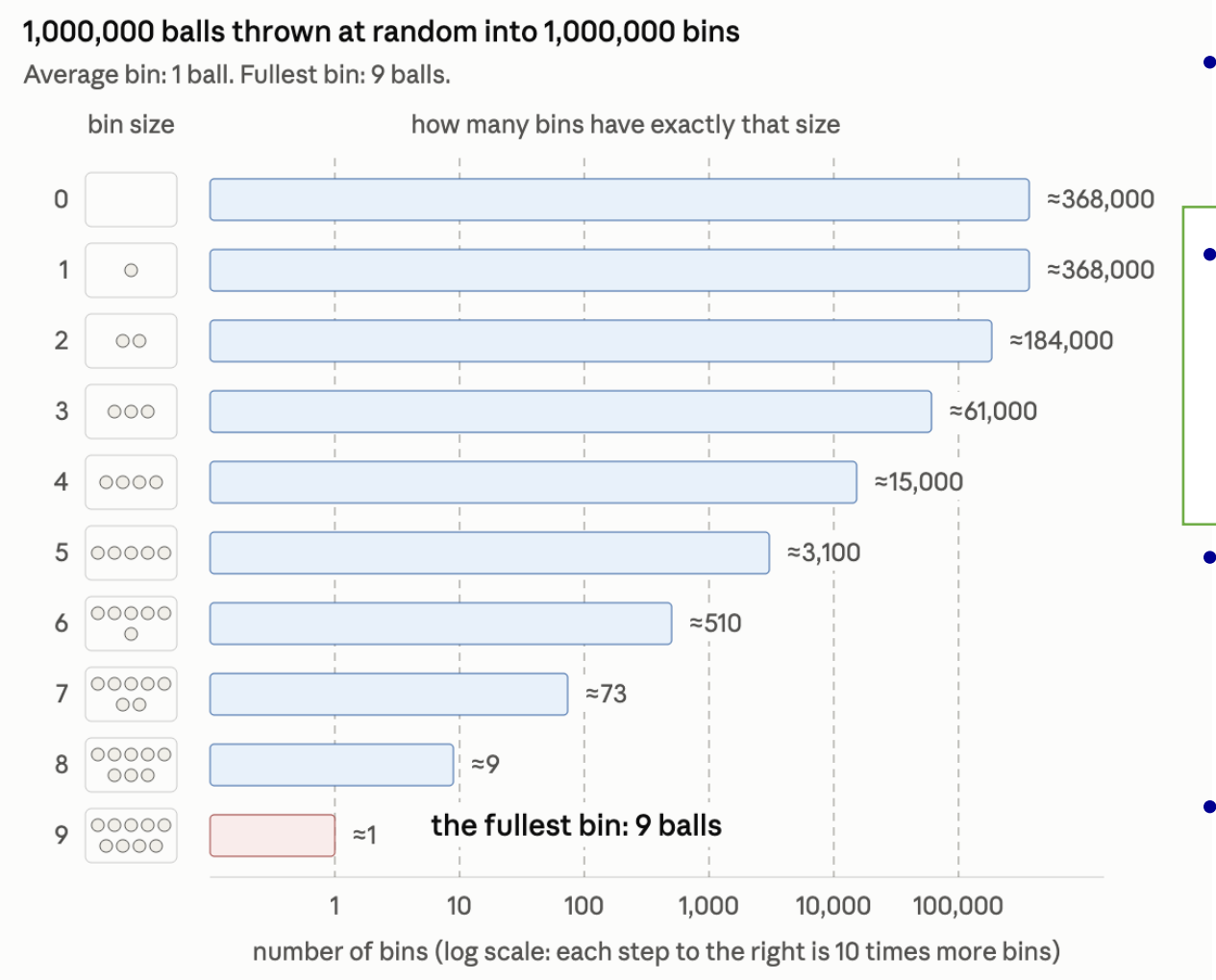


The average tells us little about the worst lookup

- Hash table with chaining. T = cost of one lookup, and $E[T] = 2$.

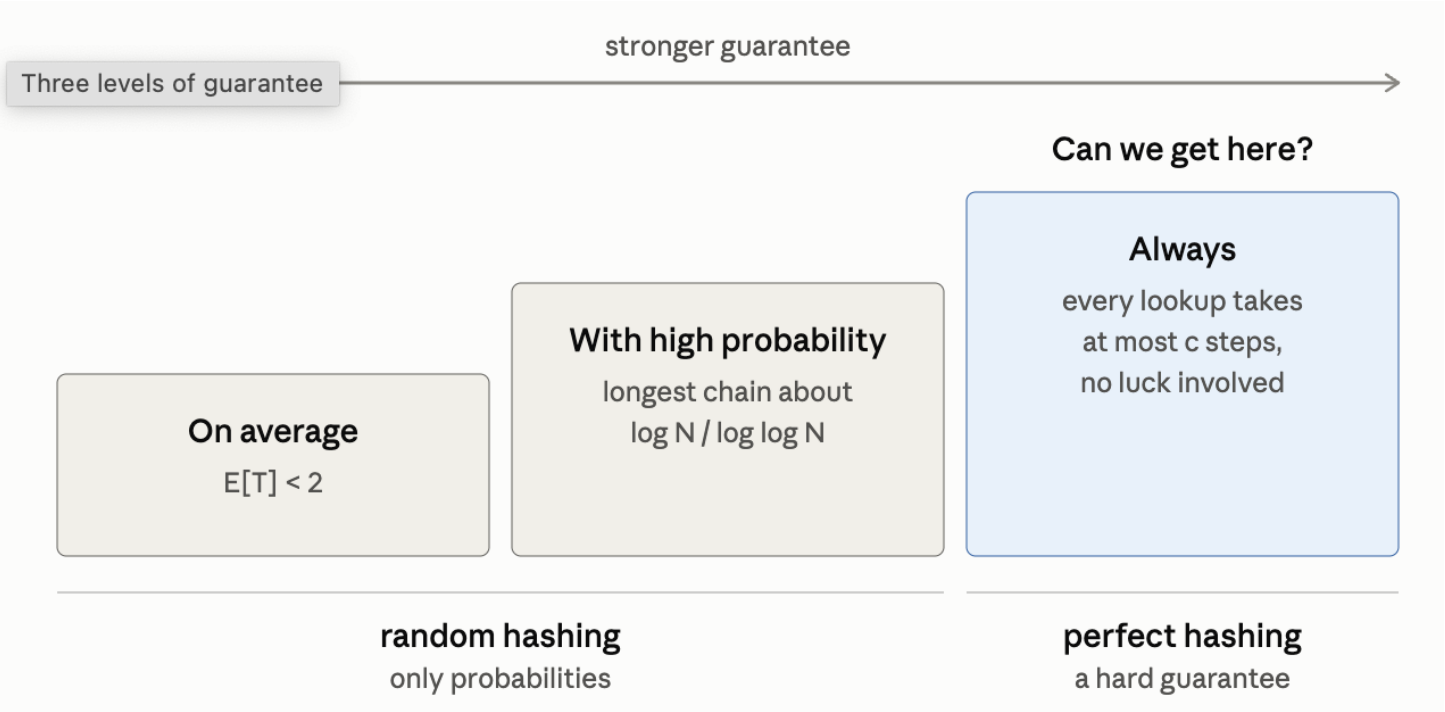


The max load is predictable with high prob.



- 10^6 keys thrown at random into 10^6 bins: about 92% of the bins hold at most 2 keys, and the fullest bin holds about 9.
- In general, the fullest bin holds about k keys with $k! \approx N/e$ (growing like $\log N / \log\log N$), and with high probability within ± 1 of that. For example, it is 5, 7, 8, 9, 10 for $N = 10^3, 10^4, 10^5, 10^6, 10^7$ (median of 20 simulations).
- This is still not a guarantee: for a given h, all N keys could land in one bin. It is improbable (probability $N^{-(N-1)}$ for fully random placement), but it is not impossible.
- Nothing here says "every lookup takes at most k steps".

Can we ever get a hard guarantee?



- **Idea**: the keys are fixed. Look at them first, choose a hash function for this set, and check that it works.
- *Randomness does not disappear*. It moves from lookup time to build time: draw h, test it, retry if it fails. After that, no probability is left in any lookup.
- First step (the pre-session): with N^2 slots it is easy, and the proof is an averaging argument. The lecture asks how to do it with only $O(N)$ slots.

Solve the set membership problem

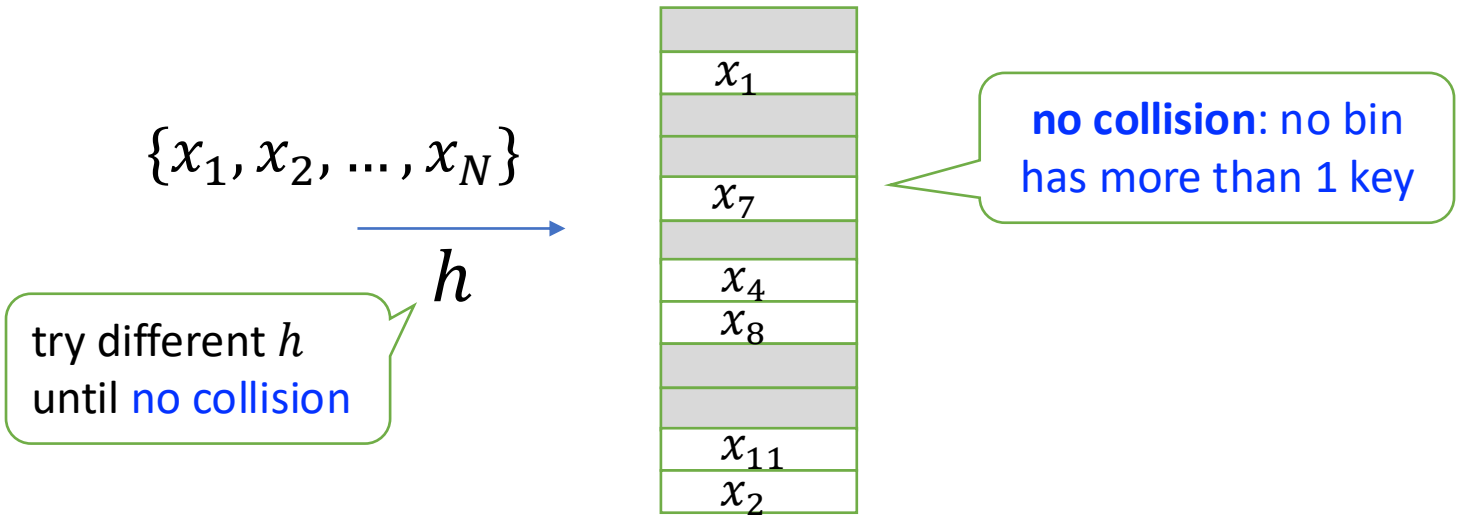
- Assume the set S of N keys is fixed.
- Given any key x, answer in $O(1)$ time if $x \in S$.
  - Same as making sure that if we use a hash table, all chain lengths are at most a fixed number, say **one**.
- How to "take out" the random length in chain formation?

for this specific set of keys!

Can I do have no collisions using space N^2?

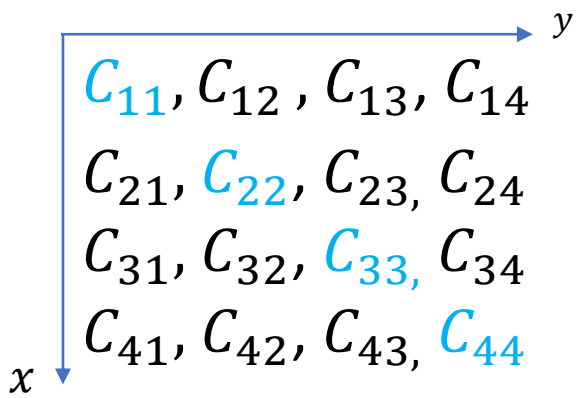
lots of space!

- Given N keys, can we construct a hash table of size $M = N^2$ with no collisions?
- Yes! by finding **the right hash function** from a set H of universal hash functions.
- **Claim**: Pick at random a hash function h. The chance of no collisions is $> \frac{1}{2}$.
- *Try different ones until no collision*! We prove that on the average we need 2 trials!

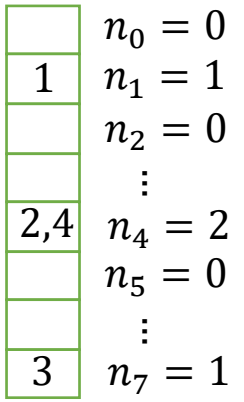


Indicator variables for collisions

- $C_{xy} \in \{0,1\} = 1_{h(x)=h(y)}$: indicator rv for *collision* of x, y using h: going to same bin.
- $x, y \in \{1, \dots, N\}$ = set of keys.
- Example: 4 keys 1,2,3,4.



$\{1,2,3,4\} \xrightarrow{h}$



$C_{11}^{1}, C_{12}^{0}, C_{13}^{0}, C_{14}^{0}$
$C_{21}, C_{22}^{1}, C_{23}, C_{24}^{1}$
$C_{31}, C_{32}, C_{33}^{1}, C_{34}$
$C_{41}, C_{42}^{1}, C_{43}, C_{44}^{1}$

Notice: $\sum_{xy} C_{xy} = 6$

Also $\sum_i n_i^2 = 1^2 + 2^2 + 1^2 = 6$

Indicator variables for collisions

- $C_{xy} \in \{0,1\} = 1_{h(x)=h(y)}$: indicator rv for *collision* of x, y using h: going to same bin.
- $x, y \in \{1, \dots, N\}$ = set of keys, $P_h[C_{xy_{\neq x}} = 1] = 1/m$

$N(N-1)/2$ $C_{xy_{\neq x}}$

$C_{11}, C_{12}, C_{13}, C_{14}, \dots, C_{1N}$

$C_{21}, C_{22}, C_{23}, C_{24}, \dots, C_{2N}$

$C_{31}, C_{32}, C_{33}, C_{34}$

$C_{41}, C_{42}, C_{43}, C_{44}, \dots, C_{4N}$

…

$C_{N1}, C_{N2}, C_{N3}, \dots C_{NN}$

Have collision: one or more $C_{xy_{\neq x}} = 1$.

$$P_h[C_{xy_{\neq x}} = 1] = 1/m$$

$$P_h[have\ collisions] = P_h[C_{12} = 1 \cup C_{13} = 1 \cup \cdots]$$

$\leq$ sum of these probabilities = $\frac{N(N-1)}{2}\frac{1}{m}$

No collision: all $C_{xy_{\neq x}} = 0$

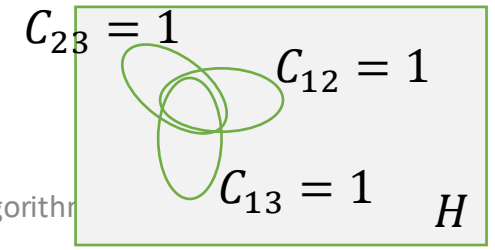
$$P_h[no\ collison] = 1 - P_h[have\ collision]$$

Proof

- **Claim**: Pick at random a hash function h. The chance of no collisions is $> \frac{1}{2}$.
  - Proof: Let $C_{xy} = 1_{h(x)=h(y)}$.
  - Note that $P_h\left[C_{xy_{\neq x}} = 1\right] \leq \frac{1}{M} = \frac{1}{N^2}$.
  - There are $\binom{N}{2} = \frac{N(N-1)}{2}$ variables $C_{xy_{\neq x}}$. % see previous slide
  - $P_h[have\ collision] = P_h[C_{xy} = 1 \cup C_{xz} = 1 \cup C_{xs} = 1 \cup \cdots]$ % at least one variable is 1

    $\leq \binom{N}{2} \frac{1}{N^2} = \frac{N(N-1)}{2} \frac{1}{N^2} < \frac{1}{2}$.

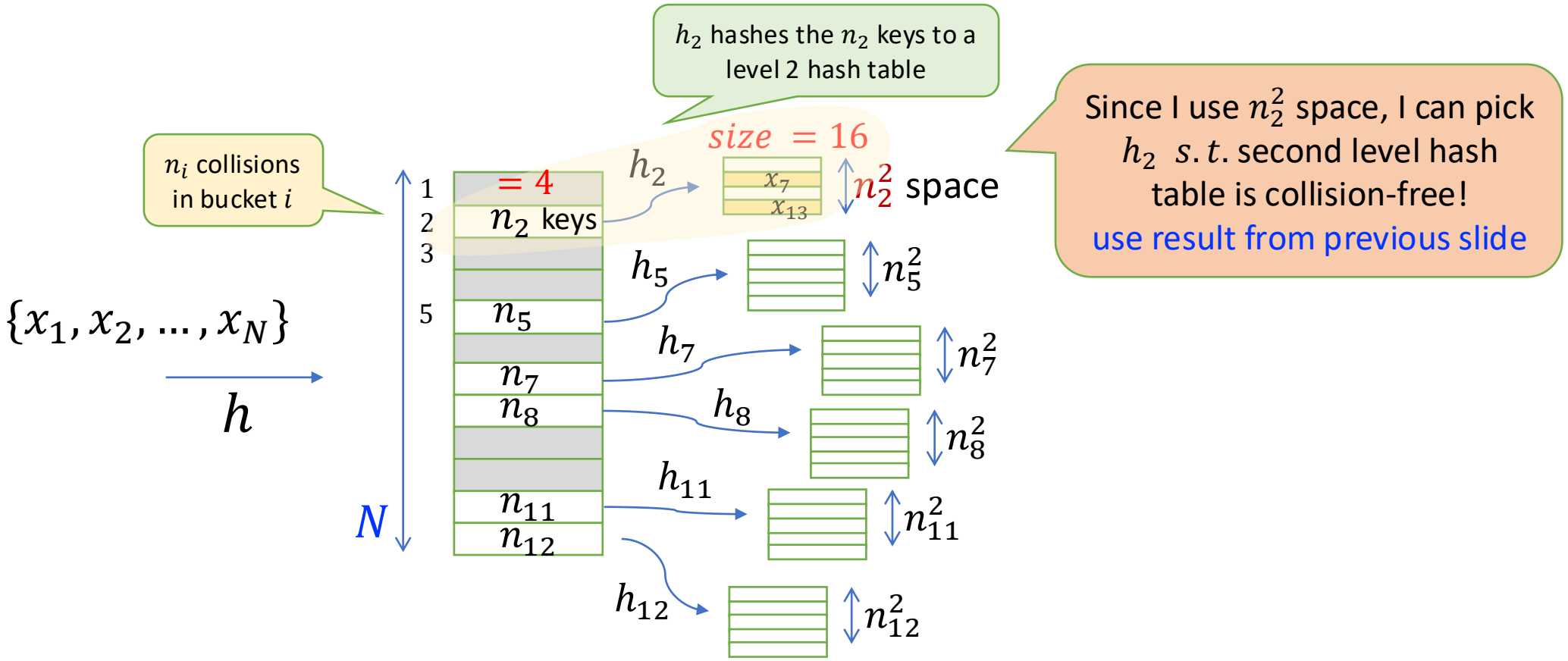
    union bound: $P(A \cup B) \leq P(A) + P(B)$
  - Hence, prob of no collisions $> \frac{1}{2}$.



$N = 3$.
$P[collision] =$
$P[C_{12} = 1 \cup C_{13} = 1 \cup C_{23} = 1]$
$\leq P[C_{12} = 1] + P[C_{13} = 1]$
$+P[C_{23} = 1]$

Can I do it using space $O(N)$?

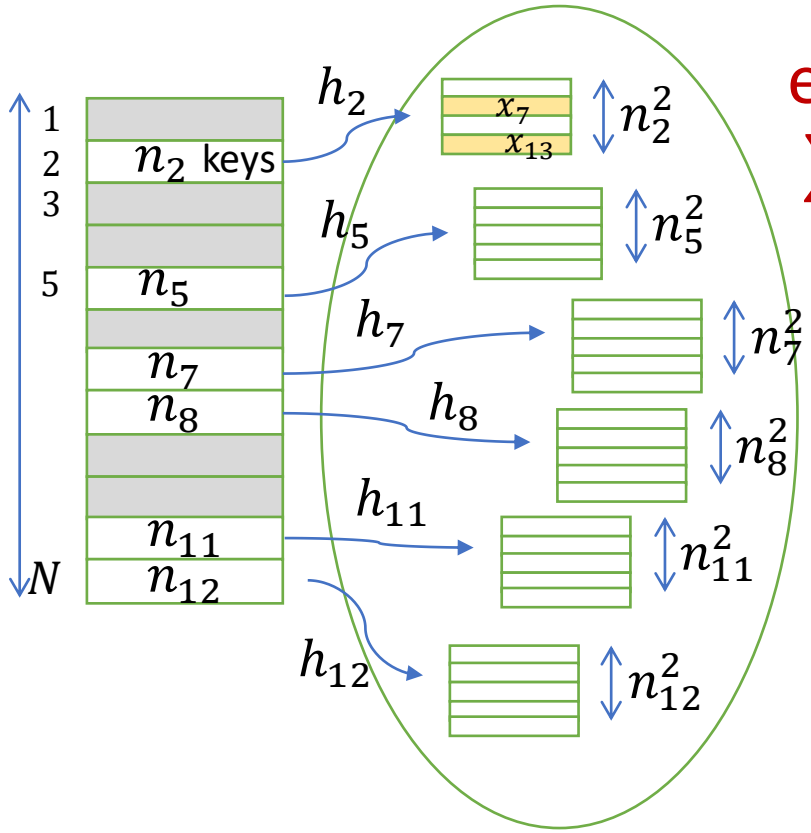
- Use a **tree of hash tables of depth 2:** no collision in level 2



Using space $O(N)$

- Use a tree of hash tables of depth 2 with no collisions in level 2.

$\{x_1, x_2, \ldots, x_N\}$ → h



Alg: keep changing h until extra space $\sum_i n_i^2 \leq 4N$

extra space $\sum_i n_i^2 \triangleq A$

We prove that $E[A] = 2N$.
Then, using Markov's inequality,

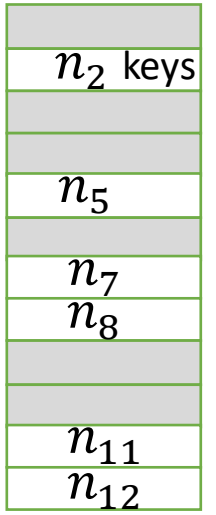
$P_h[A > 4N] \leq \frac{1}{2}$

$\Leftrightarrow P_h[A \leq 4N] > \frac{1}{2}$

It will take 2 tries of h on average to get total space $\leq 5N$.

Lemma

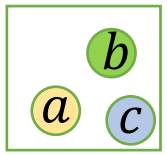
- $\sum_i n_i^2 = \sum_x \sum_y C_{xy}$.
- Proof: $\sum_x \sum_y C_{xy} = \sum_i [\sum_{x \in B_i} \sum_{y \in B_i} C_{xy}] = \sum_i n_i^2$.



a b c d

C_{aa} C_{ba}
C_{ab} C_{bb}
C_{ac} C_{bc}

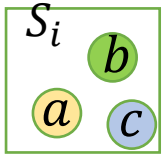
single bin case

bin i 

$B_i = \{a, b, c\}$,
$\sum_{x \in B_i} \sum_{y \in B_i} C_{xy} =$
$C_{aa} + C_{ab} + C_{ac} +$
$C_{ba} + C_{bb} + C_{bc} +$
$C_{ca} + C_{cb} + C_{cc}$
$= 9$

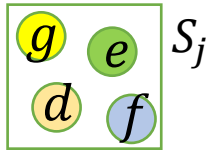
two bin case

$C_{ab} = 1, C_{ag} = 0$

B_i S_i 

$n_i = 3$
$n_i^2 = 9$

$\sum_{x \in \{a,b,c\}} \sum_{y \in \{a,b,c\}} C_{xy} = 9$

B_j 

S_j

$n_j = 4$
$n_j^2 = 16$

$\sum_{x \in \{g,e,d,f\}} \sum_{y \in \{g,e,d,f\}} C_{xy} = 16$

$C_{xy} = 0$ if x, y in different bins.

$\sum_x \sum_y C_{xy}$
$= \sum_{x \in B_i} \sum_{y \in B_i} C_{xy} + \sum_{x \in B_j} \sum_{y \in B_j} C_{xy} = n_i^2 + n_j^2$

Proof of $P_h\left[\sum_i n_i^2 \le 4N\right] > \frac{1}{2}$

- $\sum_i n_i^2 = \sum_x \sum_y C_{xy}$. % using the previous Proposition

$$\Rightarrow E_h\left[\sum_i n_i^2\right] = E_h\left[\sum_x \sum_y C_{xy}\right]$$

$$= N + E_h\left[\sum_x \sum_{y \neq x} C_{xy}\right] \quad \% \text{ since } C_{xx} = 1$$

$$= N + \sum_x \sum_{y \neq x} E_h[C_{xy}] \quad \% \text{ using linearity of expectation}$$

$$\le N + \sum_x \sum_{y \neq x} \frac{1}{M} \quad \% \text{ since } h \text{ is universal}$$

$$= N + N(N-1)\frac{1}{M_{=N}} < 2N.$$

- Markov inequality: $\Pr_h\left[\sum_i n_i^2\right] \ge 4N \le \frac{E_h\left[\sum_i n_i^2\right]}{4N} < \frac{2N}{4N} = \frac{1}{2}$.

Markov inequality:

$$\Pr[X > a] \le \frac{E[X]}{a}.$$

Conclusions

- We constructed a “static” data structure for checking membership for a given set of keys.
- There are generalizations we did not cover.
  - Easy generalization when new keys are inserted in the set
  - In this case we may need to rehash in the case of collisions in the first hash table.
  - But the amortized cost per operation is still $O(1)$.